Table 3: Comparison of Top-7 logits in `poly-context` (space), `poly-context` (save) and `poly-context` (ball). The synonyms or words relevant to the meaning of context 1 and context 2 are coloring. We also highlight the splitted token (e.g. ball-oons) in GPT-2 Small. The words that appear in both context 1 and 2 are **bolded**.

	`poly-context` (**space**)				`poly-context` (**save**)				`poly-context` (**ball**)			
	Context 1		Context 2		Context 1		Context 2		Context 1		Context 2	
rank	**tokens**	**logits**	**tokens**	**logits**	**tokens**	**logits**	**tokens**	**logits**	**tokens**	**logits**	**tokens**	**logits**
1	flight	1.224	Layout	0.894	**Save**	0.867	eLL	0.874	oons	1.170	handler	0.997
2	plane	0.957	occupied	0.882	ulative	0.696	Cache	0.744	istics	1.115	player	0.884
3	**shuttle**	0.938	spaces	0.853	luc	0.681	TOR	0.731	oon	1.026	ball	0.852
4	gravity	0.937	vacated	0.846	Saving	0.675	Marcos	0.665	antine	1.008	wright	0.842
5	craft	0.920	space	0.825	entary	0.672	Sanctuary	0.658	park	0.975	hand	0.832
6	Engineers	0.876	**shuttle**	0.799	WhatsApp	0.655	**Save**	0.649	asso	0.960	players	0.825
7	planes	0.869	occupancy	0.798	saving	0.650	orage	0.644	asted	0.905	disposal	0.814

regularization (see Appendix E). These results show that random SAE achieves the upper bound of Specificity, while densely trained SAE does not exhibit such unnatural metrics. Therefore, we basically recommend a holistic evaluation with all metrics, and, in case you would like to optimize one of them independently, we recommend the usage of class-balanced metrics such as accuracy or F1 scores as safer options.

5 Monosemantic Feature May Capture Each Meaning in Polysemy

PS-Eval is designed to evaluate SAEs through the analysis of whether the learned monosemantic features encoded from the polysemous words correspond to each word meaning, depending on the context. To clarify this core concept, we start with validating our implicit assumption of whether the SAE features encode word meanings by employing the logit lens technique (nostalgebraist, 2020).

We compute the maximum activated SAE features when two poly-context samples (Context 1, Context 2) are input. The corresponding activations for each context are calculated as:

$$a^1_{\text{max}} = W_{\text{dec}} \max(f(x_1)), \quad a^2_{\text{max}} = W_{\text{dec}} \max(f(x_2)),$$

where x_1 and x_2 represent the activations of the LLM for Context 1 and Context 2 in poly-context, respectively. Next, we apply the unembedding matrix to these activations to obtain the logits:

$$\text{logit}_1 = W_U a^1_{\text{max}}, \quad \text{logit}_2 = W_U a^2_{\text{max}}.$$

Table 3 presents the top-7 logits for the polysemous words "space", "save", and "ball" based on the above calculations. These contexts can be found in Table 6 in Appendix F. For Context 1 of the word "space", the top logits are related to terms associated with outer space (e.g., "flight", "gravity"), whereas for Context 2, the top logits are more aligned with architectural or layout-related terms (e.g., "Layout", "vacated"). This shows that the SAE features correspond to each word meaning. Furthermore, it can be observed that when the word meanings in the context are different, SAE features also reflect those distinct meanings. For the word "save," Context 1 focuses on terms related to saving in the sense of preserving or conserving resources, such as money (e.g., "Save," "Saving"), while Context 2 extracts meanings more related to saving data or technical operations (e.g., "Cache"). This distinction is clearly seen in Table 6, where Context 1 involves monetary or resource-saving, and Context 2 relates to data saving in a technical sense. Similarly, for the word "ball," Context 1 refers to a formal event or dance, where "ball" is used in the sense of a social gathering (e.g., "park"), while Context 2 reflects meanings associated with sports, where "ball" refers to a physical object used in games or sports activities (e.g., "handler," "player").

Following positive observations in prior works (Templeton et al., 2024; Cunningham et al., 2023; Li et al., 2024), these qualitative results demonstrate that a monosemantic SAE feature can correspond to a specific word meaning, which justifies our evaluation protocol.

6 Evaluating SAEs with Polysemous Words

6.1 SAE Features are More Interpretable than LLM Activations

To investigate the significance of using SAE for PS-Eval, we demonstrate that using SAE features results in better interpretability of polysemous contexts than the activations of LLMs. When a poly-

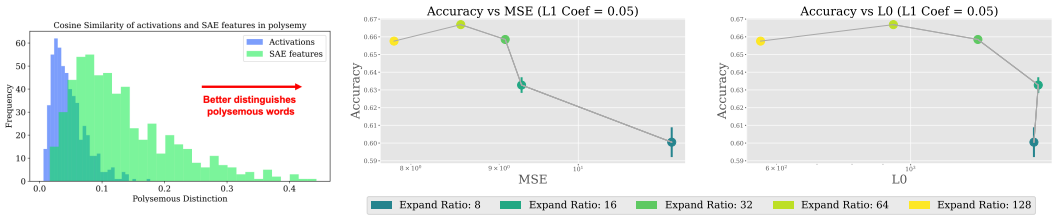

Figure 3: (**Left**) Cosine distance between LLM activations and SAE features in polysemantic contexts. The histogram compares the distribution of Polysemous Distinction (i.e. 1 - Cosine Similarity) for LLM activations (blue) and SAE features (green), demonstrating that SAE features extract more distinct representations in polysemous contexts. We adopt $R = 128$ and L_1 coefficient $\lambda = 0.05$ for the hyperparameters of SAEs. (**Right**) Comparison of Accuracy vs MSE and L_0 for different Expand Ratios (L_1 Coef = 0.05). The left panel shows the relationship between accuracy and mean squared error (MSE), while the right panel presents accuracy versus normalized L_0 sparsity. The markers represent various expand ratios (8, 16, 32, 64, and 128). Higher expand ratios generally show a trend towards better performance, as indicated by the yellow and green markers corresponding to expand ratios of 64 and 128.

semous context is input, we denote the LLM's activation for the target word as $a_{\text{LLM}}(\text{poly-context})$, and the SAE feature for the same context as $f_{\text{SAE}}(\text{poly-context})$. We calculated the cosine distance between these activations and SAE features to assess their similarity or divergence, which we refer to as *Polysemous Distinction*:

$$1-\frac{a_{\text{LLM}}(\text{poly-context}_1)\cdot a_{\text{LLM}}(\text{poly-context}_2)}{\|a_{\text{LLM}}(\text{poly-context}_1)\|\|a_{\text{LLM}}(\text{poly-context}_2)\|},\ 1-\frac{f_{\text{SAE}}(\text{poly-context}_1)\cdot f_{\text{SAE}}(\text{poly-context}_2)}{\|f_{\text{SAE}}(\text{poly-context}_1)\|\|f_{\text{SAE}}(\text{poly-context}_2)\|}.$$

As shown in Figure 3 (right), the cosine distance between the LLM activations for different polysemous contexts is small, indicating that the LLM struggles to differentiate between distinct meanings in polysemous contexts (i.e., less interpretable activations). In contrast, SAE features show a larger cosine distance, effectively capturing the differences between the contexts with different meanings. This result demonstrates that SAE effectively disentangles different meanings, and, for polysemous words, it enhances the interpretability of internal representations in LLMs.

6.2 Increasing Expand Factor Improves Polysemous SAE Feature

Previous works (Templeton et al., 2024; Gao et al., 2024) have reported that increasing the dimensionality of SAE features leads to better performance in terms of MSE and L_0 metrics, and we may also expect that it allows for learning more fine-grained features from activations (Bricken et al., 2023). To investigate the relationship between PS-Eval performance and the scaling of latent variables, we evaluated the performance of our model trained with various expansion ratios (defined in Section 3.1) in Figure 3 (right). Similar to MSE and L_0, we observed that increasing the expand ratio led to higher accuracy on PS-Eval. This trend indicates that a higher expand ratio enables the model to better extract the monosemantic fetures from polysemantic activations. However, we found that the accuracy saturates around an expand ratio of 64, suggesting that there is an optimal latent dimension for practical SAE usage. Here, we define L_0 as the number of activated SAE features. See Appendix G for other metrics.

6.3 Alternative activation functions to ReLU are not always better

Various activation functions for SAE have been proposed to achieve a better Pareto optimality between MSE and L_0 sparsity. More recent work has introduced TopK, which replaces $\text{ReLU}(x)$ with $\text{TopK}(\text{ReLU}(x))$ (Gao et al., 2024), and JumpReLU, which modifies the threshold of the ReLU activation to improve performance in terms of both MSE and sparsity (Rajamanoharan et al., 2024b) (see Appendix H for the detailed descriptions).

We here evaluate whether these alternative activation functions also yield better results when applied to our evaluation metrics. In Figure 4, we compare models trained with TopK activations, where k is varied over values such as $k = \{384, 192, 96\}$, and models trained with JumpReLU, including its STE variant. For JumpReLU, the threshold is adjusted over the range of $[0.0001, 0.001]$, while for the STE variant, the threshold represents an initial value. We note that TopK typically uses half the dimension of the model as the value of k (e.g., for GPT-2 small, this corresponds to $768/2 =$

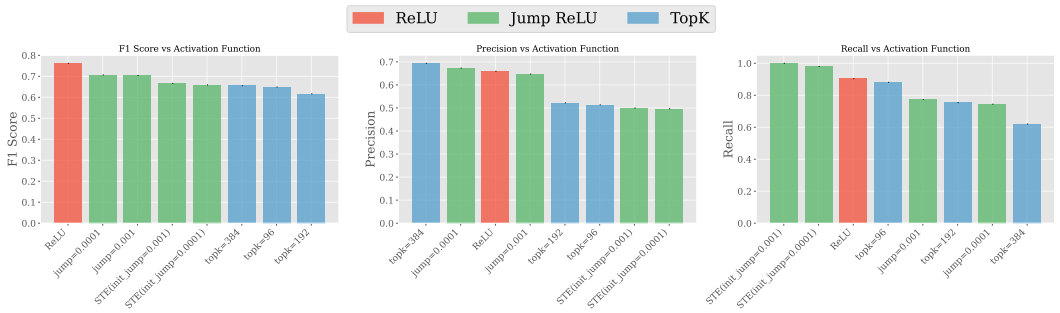

Figure 4: Comparison of F1 score, Precision, and Recall across different activation functions. The models were trained with JumpReLU, including its STE variant (with varying thresholds), TopK (with different k), and ReLU. The results imply that ReLU may be still better than JumpReLU or TopK with the polysemous words.

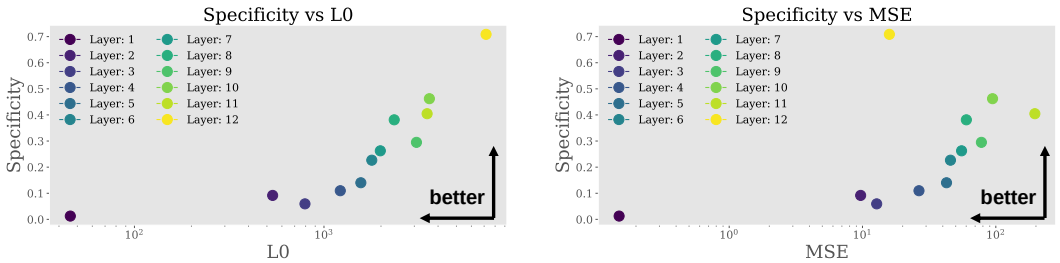

Figure 5: Comparison of Specificity across different layers of the SAE in relation to L_0 (left) and MSE (right). Each point represents a layer, from layer 1 to layer 12, with deeper layers (i.e., layer 6 onwards) showing an increasing trend in Specificity. This indicates that deeper layers exhibit more distinct activations for polysemous words, consistent with previous findings that deeper layers in SAEs face greater challenges in training, leading to higher MSE and L_0 scores.

384), and that JumpReLU often adopts a threshold of 0.001 in the original works (Gao et al., 2024; Rajamanoharan et al., 2024b).

Figure 4 compares these activation functions in terms of F1 score, Precision, and Recall. ReLU achieves the highest F1 score, with the order being ReLU > JumpReLU > TopK. For TopK, smaller k leads to higher Recall, and JumpReLU-STE further boosts Recall. These results suggest that focusing solely on the trade-off between MSE and L_0 may not effectively enhance semantic quality.

7 Locating Polysemy Circuits in LLMs

Polysemy is Captured in Deeper Layers We investigate where large language models (LLMs) interpret polysemous words differently by using Specificity (calculated as Specificity $= \frac{TN}{TN+FP}$) based on the confusion matrix from PS-Eval (see Table 2). Specificity allows us to assess whether the features in the SAE activate differently when presented with Poly-context (contexts involving words with defferent meanings).

Figure 5 shows the Specificity of each SAE layer in relation to L_0 and MSE. Previous studies (Gao et al., 2024) have demonstrated that training SAEs becomes increasingly difficult as the depth of the layers increases, which is consistent with our findings, as both MSE and L_0 rise in deeper layers. While these two metrics increase, we observe that Specificity improves around the 6th layer, indicating that polysemous words are more easily decomposed into monosemantic features through SAE training. These results demonstrate an inverse relationship between MSE/L_0 and Specificity, suggesting that focusing solely on improving the Pareto frontier of MSE or L_0, as emphasized in prior research, may overlook important insights.

SAE Features Vary by Transformer Component While early prior work (Bricken et al., 2023) primarily centered on the activations within MLP layers, more recent studies have broadened this scope not only to MLP layers but also to residual and attention layers (Kissane et al., 2024; He et al., 2024). We explore which components of a single Transformer layer contribute to the acquisition of monosemantic features from polysemous words. Specifically, we divided the activations from a single Transformer layer into three distinct outputs: Residual, MLP, and Attention. See Appendix I